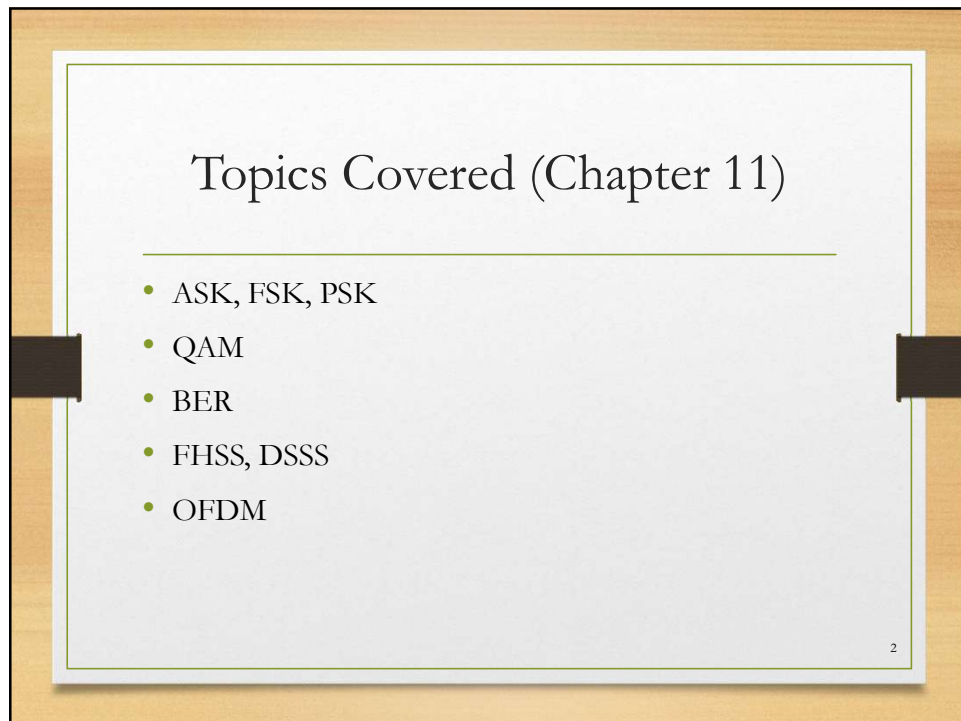




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11-4: Basic Modem Concepts

- Digital data are transmitted over the telephone and cable television networks by using broadband communication techniques involving **modulation**, which are implemented by a **modem**, a device containing both a **modulator** and a **demodulator**.
- Modems convert binary signals to analog signals capable of being transmitted over telephone and cable TV lines and by radio, and then demodulate such analog signals, reconstructing the equivalent binary output.

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11-4: Basic Modem Concepts

- There are four widely used modem types:
 1. Conventional analog dial-up modems.
 2. Digital subscriber line (DSL) modems.
 3. Cable TV modems.
 4. Wireless modems.

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11-4: Basic Modem Concepts

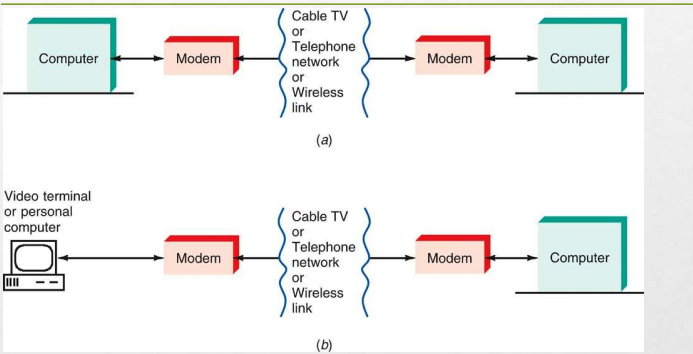


Figure 11-12: How modems permit digital data transmission on the telephone network.

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11-4: Basic Modem Concepts

Modulation for Data Communication

- The five main types of modulation used in modern modems are:
 1. Amplitude-Shift Keying (ASK)
 2. Frequency-Shift keying (FSK)
 3. Phase-Shift Keying (PSK)
 4. Quadrature Amplitude Modulation (QAM)
 5. Orthogonal Frequency Division Multiplexing (OFDM)

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3-3: Sidebands and the Frequency Domain

- When complex signals such as pulses or rectangular waves modulate a carrier, a broad spectrum of sidebands is produced.
- A modulating square wave will produce sidebands based on the fundamental sine wave as well as the third, fifth, seventh, etc. harmonics.
- Amplitude modulation by square waves or rectangular pulses is referred to as **amplitude shift keying (ASK)**.
- ASK is used in some types of data communications.

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3-3: Sidebands and the Frequency Domain

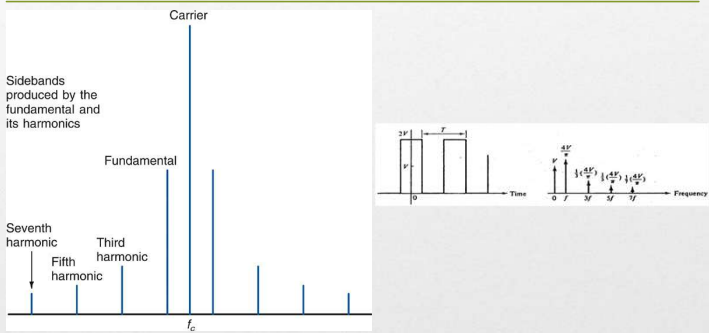
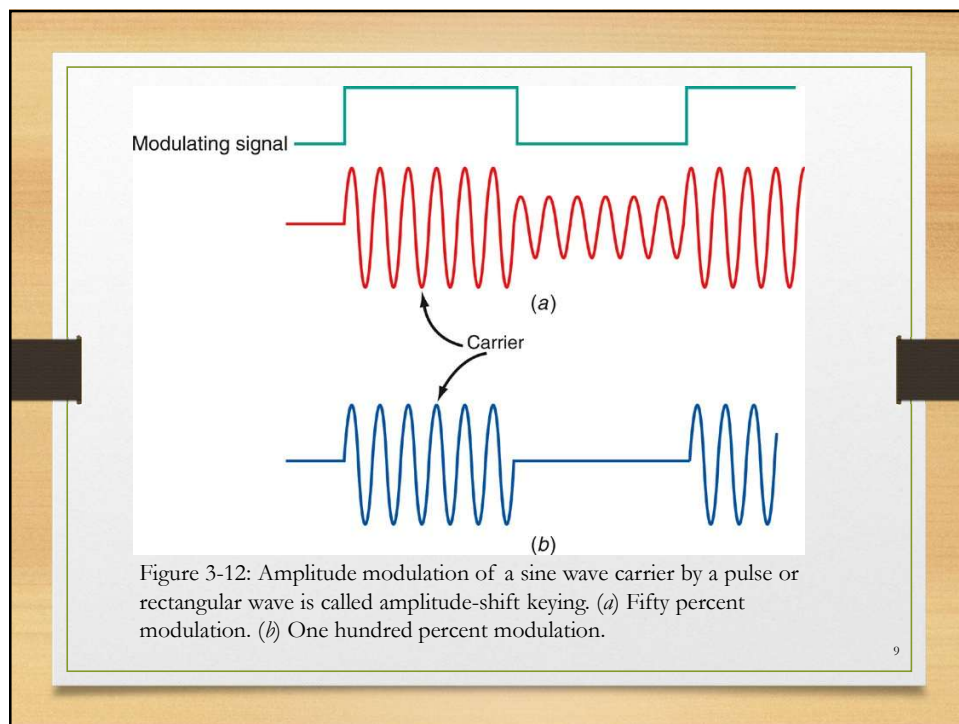


Figure 3-11: Frequency spectrum of an AM signal modulated by a square wave.

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3-3: Sidebands and the Frequency Domain

- **Continuous-wave (CW)** transmission can be achieved by turning the carrier off and on, as in Morse code transmission.
- Continuous wave (CW) transmission is sometimes referred to as **On-Off keying (OOK)**.
- **Splatter** is a term used to describe harmonic sideband interference.

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Frequency-Shift Keying (FSK)

- **Frequency-shift keying (FSK)** is the oldest and simplest form of modulation used in modems.
- In FSK, two sine-wave frequencies are used to represent binary 0s and 1s.
- For example, a binary 0 and 1 (called a **space** and **mark**) has a frequency of 1070 Hz, and 1270 Hz, respectively.
- These two frequencies are alternately transmitted to create the serial binary data.

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Frequency-Shift Keying (FSK)

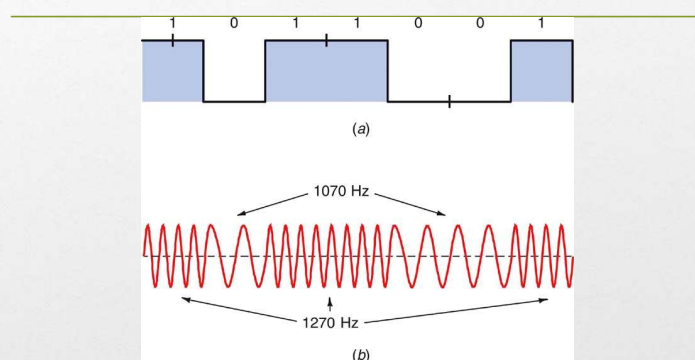


Figure 11-13: Frequency-shift keying. (a) Binary signal. (b) FSK signal.

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Phase-Shift Keying

- In **phase-shift keying (PSK)**, the binary signal to be transmitted changes the phase shift of a sine-wave character depending upon whether a binary 0 or binary 1 is to be transmitted.
- A phase shift of 180° , the maximum phase difference that can occur, is known as a **phase reversal**, or **phase inversion**.
- During the time that a binary 0 occurs, the carrier is transmitted with one phase; when a binary 1 occurs, the carrier is transmitted with a 180° phase shift.

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Phase-Shift Keying

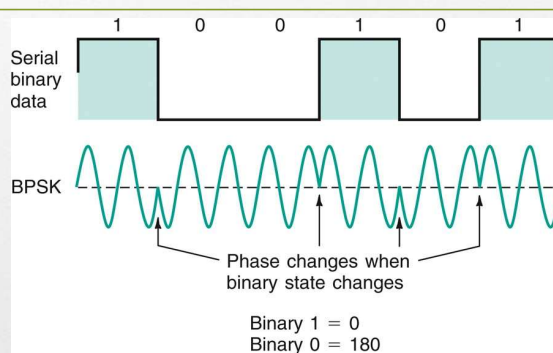


Figure 11-18: Binary phase-shift keying (BPSK).

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QPSK

- One way to increase the binary data rate while not increasing the bandwidth required for signal transmission is to encode more than 1 bit per phase change, so that a symbol change (phase shift) represents multiple bits.
- One commonly used system is the **quadrature, quaternary, or quadra phase PSK (QPSK or 4-PSK)**, where each pair of serial digital bits, called a **dibit**, is assigned a particular phase.

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QPSK

Dibit		Phase shift
0	0	45
0	1	135
1	1	225
1	0	315

135 = 01 45 = 00

225 = 11 315 = 10

90°

180° 0°

270°

(a) (b) (c)

Figure 11-24: Quadrature PSK modulation. (a) Phase angle of carrier for different pairs of bits. (b) Phasor representation of carrier sine wave. (c) Constellation diagram of QPSK.

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QPSK

- The modulator has a 2-bit shift register implemented with flip-flops, commonly known as a **bit splitter**.
- Serial binary data train is shifted through register, and bits from the flip-flops are applied to balanced modulators.
- Carrier oscillator is applied to one balanced modulator and a 90° phase-shifted version to another balanced modulator.
- Outputs of the balanced modulators are linearly mixed to produce the QPSK signal.

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QPSK

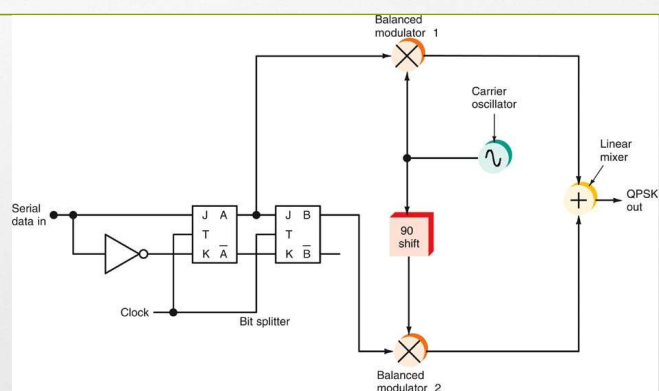


Figure 11-25: A QPSK modulator.

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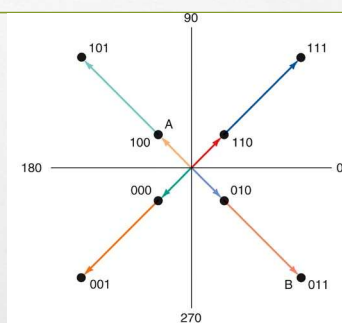
QAM

- One of the most popular modulation techniques used in modems for increasing the number of bits per baud is **quadrature amplitude modulation (QAM)**.
- QAM uses both amplitude and phase modulation of carrier.
- In 8-QAM, there are 4 phase shifts and 2 carrier amplitudes, resulting in 8 possible states.
- With eight states, 3 bits can be encoded for each baud or symbol transmitted.
- Each 3-bit binary word transmitted uses a different phase-amplitude combination.

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QAM



Note: Each vector has a specific amplitude and phase shift and represents one 3-bit word.

Figure 11-29: A constellation diagram of a QAM signal.

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Spectral Efficiency

- Spectral efficiency is a measure of how fast data can be transmitted in a given bandwidth (bps/Hz).
- Different modulation methods give different efficiencies.

Modulation	Spectral efficiency, bps/Hz
FSK	<1
GMSK	1.35
BPSK	1
QPSK	2
8-PSK	3
16-QAM	4

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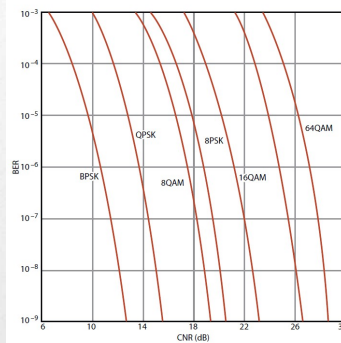
BER

- The signal-to-noise (S/N) ratio clearly influences the spectral efficiency.
- The greater the noise, the greater the number of bit errors.
- The number of errors that occur in a given time is called the **bit error rate (BER)**.
- The BER is the ratio of the number of errors that occur to the number of bits that occur in a one second interval.

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BER



7 This is a comparison of several popular modulation methods and their spectral efficiency expressed in terms of BER versus CNR. Note that for a given BER, a greater CNR is needed for the higher QAM levels.

$$\frac{E_b}{N_0} = \left(\frac{C}{N} \right) \left(\frac{B}{R_b} \right)$$

For BPSK, the BER can be calculated as

$$BER = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right)$$

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-t^2} dt$$

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11-5: Wideband Modulation

- While most modulation methods are designed to be spectrally efficient, there is another class of modulation methods that does just the opposite.
- These methods are designed to use more bandwidth. The transmitted signal occupies a bandwidth many times greater than the information bandwidth.
- The two most widely used wideband modulation methods are **spread spectrum** and **orthogonal frequency-division multiplexing**.

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Spread Spectrum

- **Spread spectrum (SS)** is a modulation and multiplexing technique that distributes a signal and its sidebands over a very wide bandwidth.
- After World War II, spread spectrum was developed by the military because it is a secure communication technique essentially immune to jamming.
- Since mid-1980s, spread spectrum has been used in a variety of commercial communication systems.

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Spread Spectrum

- The most widespread use was in cell phones, referred to as **code-division multiple access (CDMA)**.
- Two basic types of spread spectrum: **frequency-hopping spread spectrum (FHSS)** and **direct-sequence spread spectrum (DSSS)**.

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FHSS

- In a **FHSS transmitter**, the serial binary data to be transmitted is applied to a conventional two-tone FSK modulator.
- The modulator output is applied to a mixer, which is also driven by a frequency synthesizer.
- The frequency synthesizer is in turn driven by a **pseudorandom code generator**, which is either a special digital circuit or the output of a microprocessor.
- The output signal from the mixer after the bandpass filter is the **difference** between one of the two FSK waves and the signal of the frequency synthesizer.

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FHSS

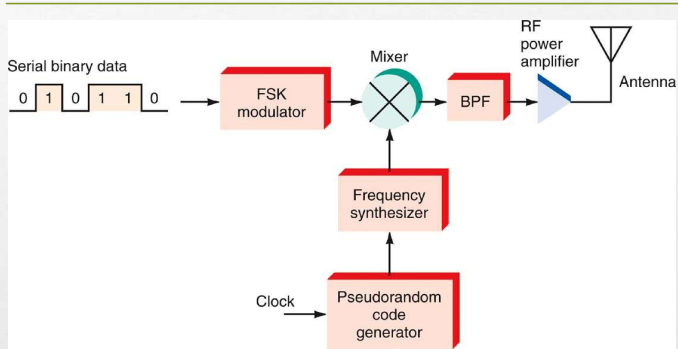


Figure 11-33: A FHSS transmitter.

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FHSS

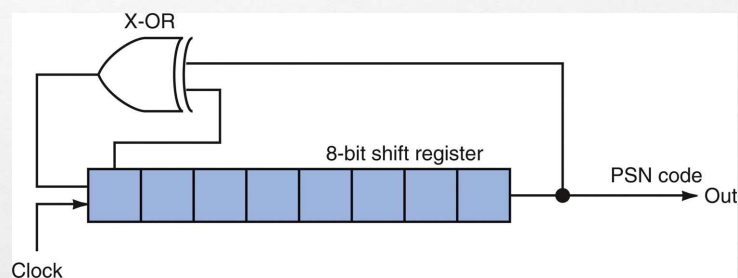


Figure 11-34: A typical pseudorandom noise (PSN) code generator.

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FHSS

- In a FHSS system, the rate of synthesizer frequency change can be higher than the data rate.
- This means that the frequency synthesizer can switch frequency multiple times during one data interval.
- The time the synthesizer remains on a single frequency is called the **dwell time**, during which the mixer creates a new carrier frequency.
- The resulting signal, whose frequency hops around, effectively scatters pieces of the signal all over the band.

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DSSS

- In a **DSSS transmitter**, the serial binary data along with a serial pseudorandom code are applied to an X-OR gate.
- One bit time for the pseudorandom code is called a **chip**, and the rate of the code is called **chipping rate**, which is faster than the data rate.
- X-OR output is then applied to a PSK modulator (some form of balanced modulator), typically a BPSK device.
- The carrier phase is switched between 0 and 180° by the 1s and 0s of the X-OR output.

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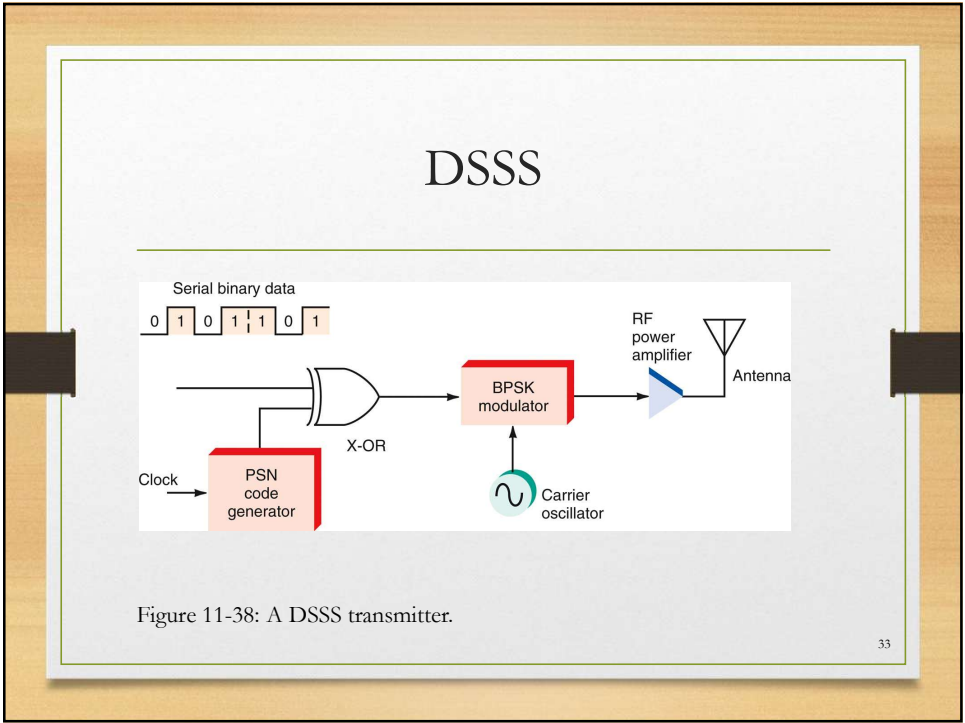
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DSSS

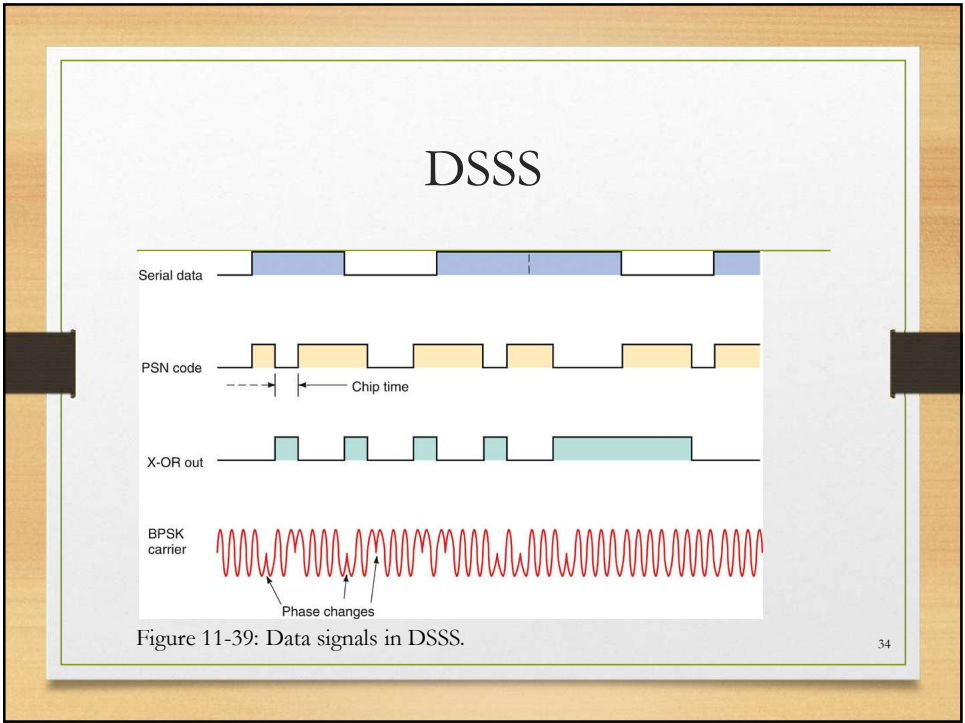
- The signal phase modulating the carrier, being much higher in frequency than the data signal, causes the modulator to produce multiple widely spaced sidebands such that the complete signal takes up a great deal of the spectrum. Thus, the resulting signal is spread.
- Because of its randomness, the signal looks like wideband noise to a conventional narrowband receiver.

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DSSS

- DSSS is also called **code-division multiple access (CDMA)** or **SS multiple access**.
- **Multiple access** refers to any technique that is used for multiplexing many signals on a single communication channel.
- CDMA is used in satellite systems so that many signals can use the same transponder.
- It is also widely used in cell phone systems. It permits more users to occupy a given band than other methods.

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Benefits of Spread Spectrum

- Security: SS prevents unauthorized listening.
- Resistance to jamming and interference: Jamming signals are typically restricted to a single frequency, and jamming one frequency does not interfere with an SS signal.
- Band sharing: Many users can share a single band with little or no interference.

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Benefits of Spread Spectrum

- Resistance to fading and multipath propagation: SS virtually eliminates wide variations of signal strength due to reflections and other phenomena during propagation.
- Precise timing: Using pseudorandom code in SS provides receiver with a way to precisely determine start/end of a transmission, making it a superior method for applications that need accurate knowledge of transmission time.

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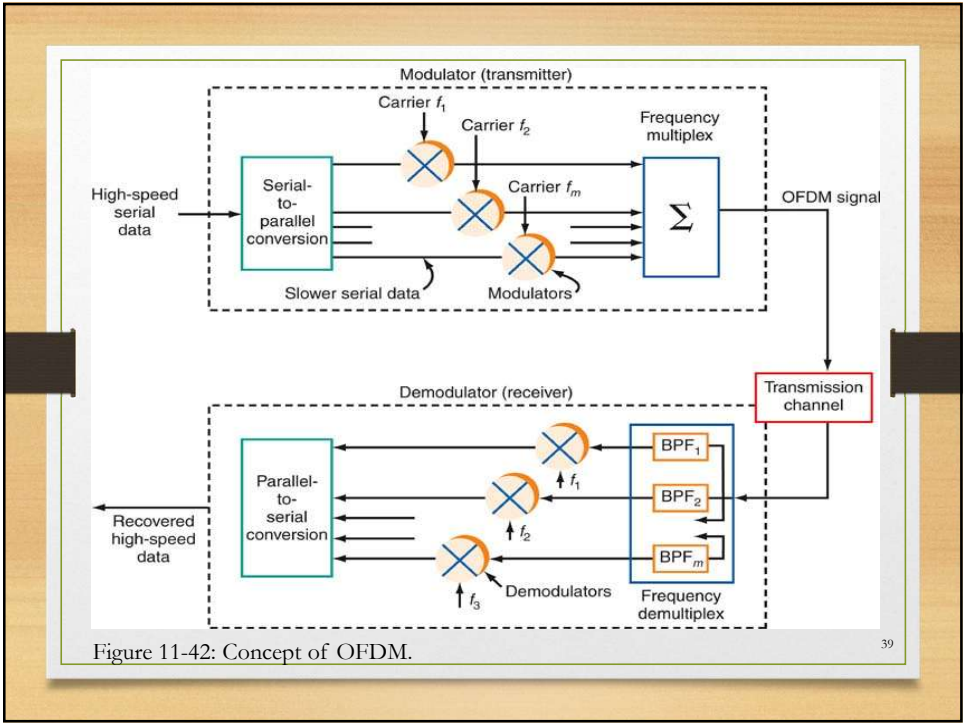
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OFDM

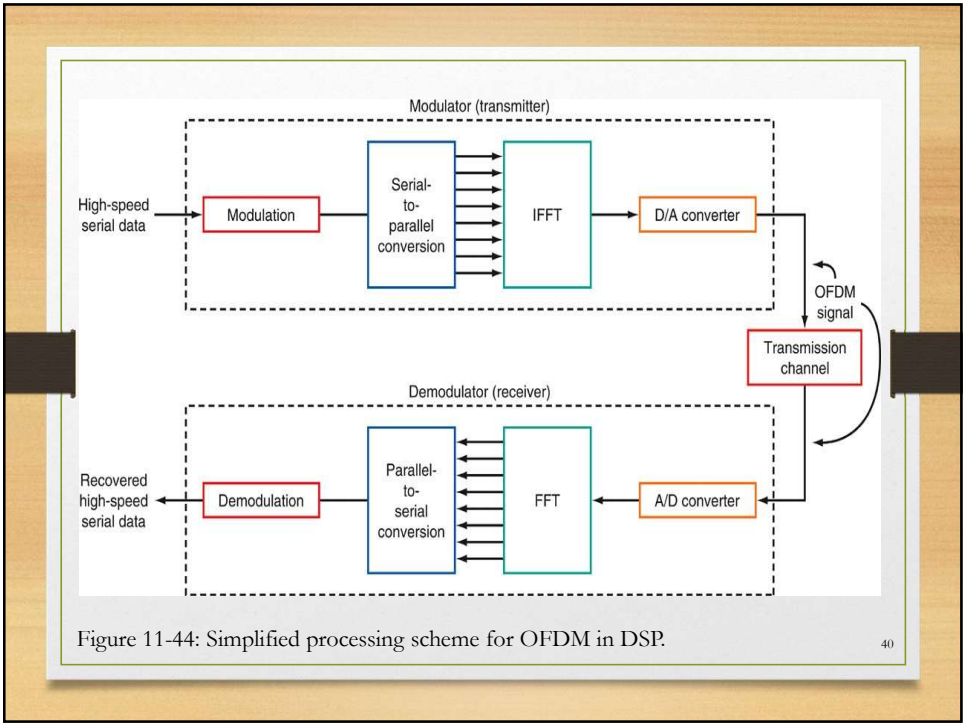
- Another wideband modulation method, also known as **multicarrier modulation**.
- Transmits data by modulating high-speed serial bit stream onto multiple orthogonal carriers, which are then frequency-multiplexed in the channel.
- This technique spreads the signals over a wide bandwidth, making them less sensitive to the noise, fading, reflections, and multipath transmission effects common in microwave communication.

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